Diagnosing Causation Claims and Biased Study Designs

Check the arithmetic first, then say what is wrong with the claim.

Directions:

- Each problem reports a real-sounding claim. In every case the numbers are correct and the conclusion is not.
- Part (a) asks you to compute the figure the claim rests on. The later part asks you to name the flaw in a sentence or two.
- Give exact fractions in lowest terms unless a decimal is requested.

1. A press release says: "Sales of our new flavour *doubled* this month — up 100%!" The shop sold 4 units last month and 8 this month, out of 5000 units of all flavours sold this month.

(a) What fraction of this month's total sales was the new flavour? _________

(b) Explain why the headline is misleading even though the arithmetic is right.

2. A club with 3000 members posts a survey on its website. Of the 240 members who replied, 210 said they like the new logo. The club announces: "Seven-eighths of our members like the new logo."

(a) What proportion of the *respondents* said they like it? _________

(b) Explain why that proportion cannot be reported as a fact about all 3000 members.

3. Across the summer months, a town's weekly ice cream sales and its weekly count of swimming-pool rescues have correlation $r = 0.90$. A blogger concludes that eating ice cream makes people need rescuing.

(a) Compute r^2 for these data. _________

(b) Name the flaw in the blogger's causal claim.

4. A reporter stands outside a gym and asks people leaving it whether they exercise daily. Of the 60 people asked, 45 say yes. The story runs as “Three-quarters of townspeople exercise every day”.

(a) What proportion of the people asked said yes? _________

(b) Name the bias in this sample and say which way it pushes the result.

5. Five students who chose to take a test-prep course scored $78, 80, 82, 84, 86$. Five who did not take it scored $70, 72, 74, 76, 78$. The course advertises: “Our students score 8 points higher.”

(a) Find the mean score of the course group. _________

(b) Find the mean score of the other group. _________

(c) Explain why this 8-point gap does not show that the course works, and describe the study that would.

6. A drug trial reports: "Our drug cuts the risk of the disease by 50%." In the trial the risk fell from 2 cases per 1000 people to 1 case per 1000 people.

(a) Compute the absolute change in risk, as an exact fraction. _________

(b) Explain why the 50% headline gives a misleading impression of the benefit.

7. A gardener tests a fertilizer on a field of 400 plants. She treats the 200 plants on the sunny south half and leaves the 200 on the shaded north half untreated. Of the treated plants 150 thrived; of the untreated plants 100 thrived. She concludes the fertilizer works.

(a) What proportion of the treated plants thrived? _________

(b) Name the flaw in this design and describe how to fix it.

8. Synthesis. A city reports hospital admissions of Nov 120, Dec 210, Jan 260, Feb 190. Flu-shot campaigns run over the same four months, and a commentator writes: "Admissions rise exactly when flu shots are given — the shots are sending people to hospital."

(a) Find the total admissions over the four months. __________

(b) Find the admissions in the busiest month. __________

(c) Explain what is wrong with the commentator's reasoning, and state what evidence would be needed to settle the question.